Peak Shaving Optimization at Columbia University

Techno-Economic Analysis of Energy Storage Systems

Presenters: Arpit Patel & Himangi Maheshwari

Faculty Advisor: Bolun Xu

Department: Earth and Environmental Engineering

Project Overview & Objectives

Objective: To reduce Columbia University's electricity costs by lowering peak demand charges through optimal energy storage scheduling.

Deterministic Baseline: Perfect Foresight

Calculate the "perfect world" savings assuming 100% accurate knowledge of future demand.

Predictive Modeling: ML Forecasts

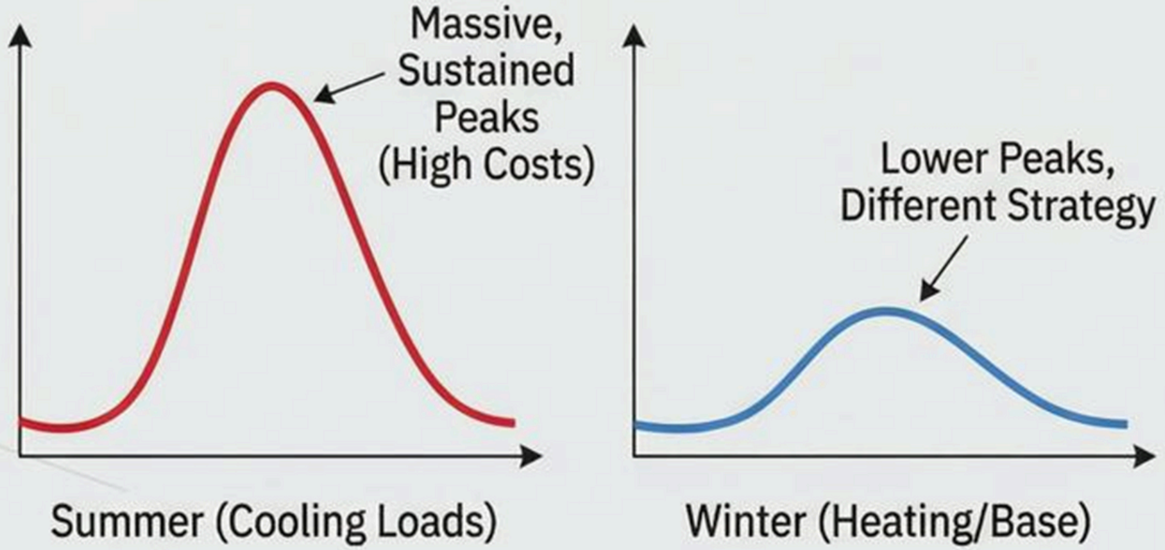
Develop Machine Learning models to forecast 2019 demand using 2018 data.

Optimization under Uncertainty: Real-World Simulation

Simulate real-world battery operation using our forecasts to determine financial viability.

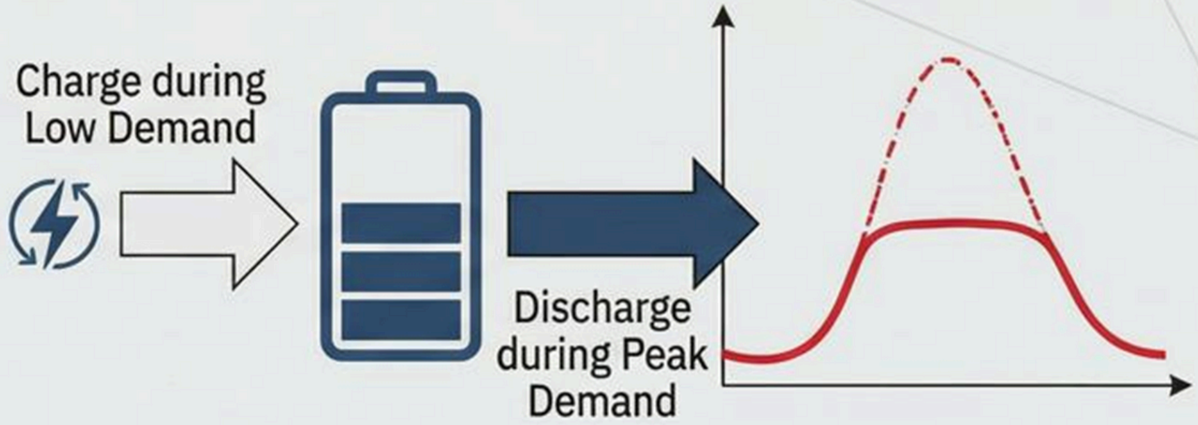
The Challenge: Peak Demand

Context & Seasonal Profiles



Columbia University faces significant electricity costs driven by Peak Demand Charges, based on the highest power usage interval in a month. High summer cooling creates sustained peaks, while winter peaks are lower and less frequent.

Goal: Peak Shaving Strategy



Discharge stored energy during high-demand periods to "shave" the peaks and reduce charges.

Implication: The energy storage system must be flexible enough to handle both distinct seasonal profiles effectively.

Stage 1 - Deterministic Optimization

Baseline Analysis & Efficiency Trade-off

Lithium-Ion Batteries	Thermal Energy Storage
Efficiency	Cost of Efficiency Loss
Efficiency: 95%	Efficiency: 70%
Cost of Loss: $0.014/kWh	Cost of Loss: $0.135/kWh

Perfect Foresight Assumption

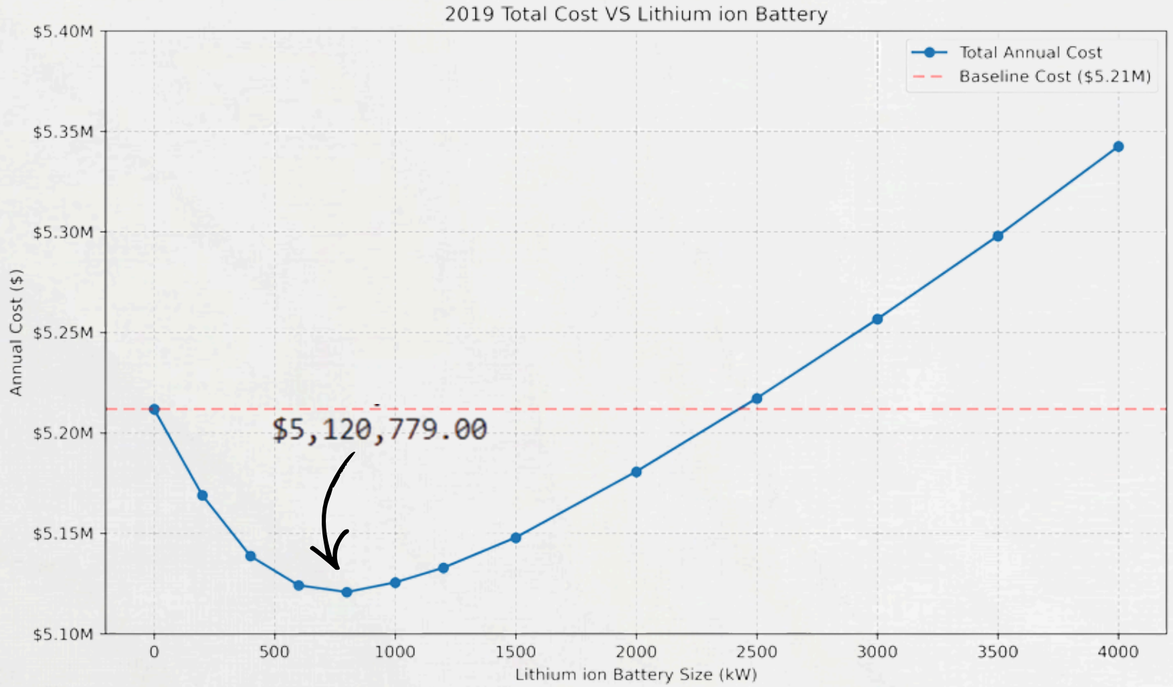
The Result: Optimal Solution & Key Metrics

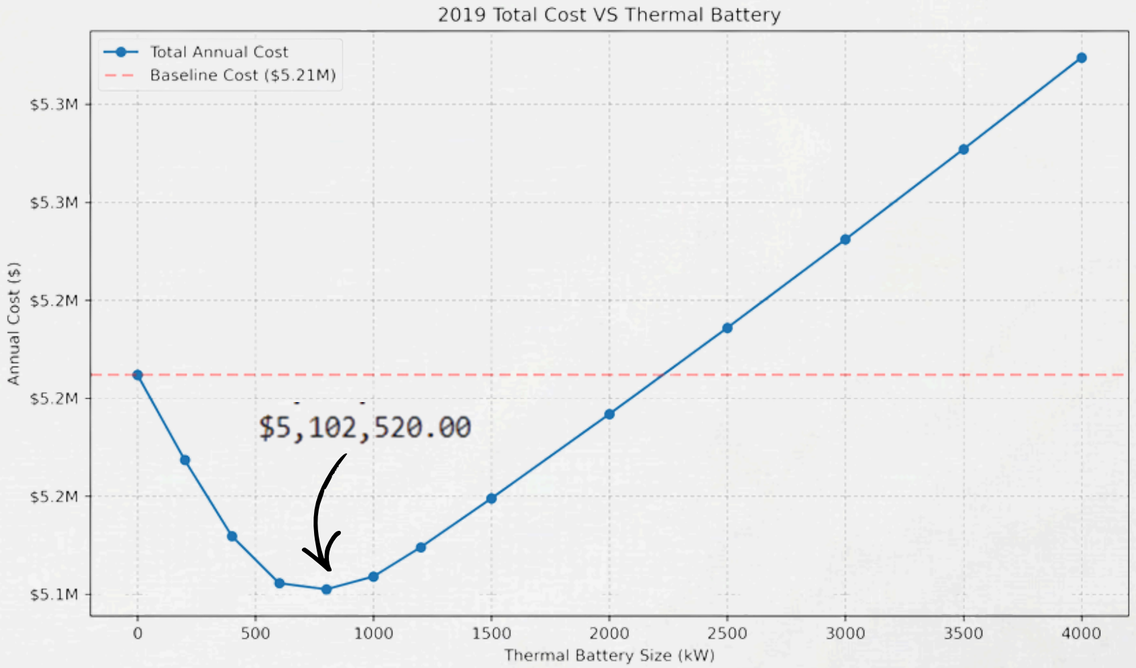
Thermal Energy Storage

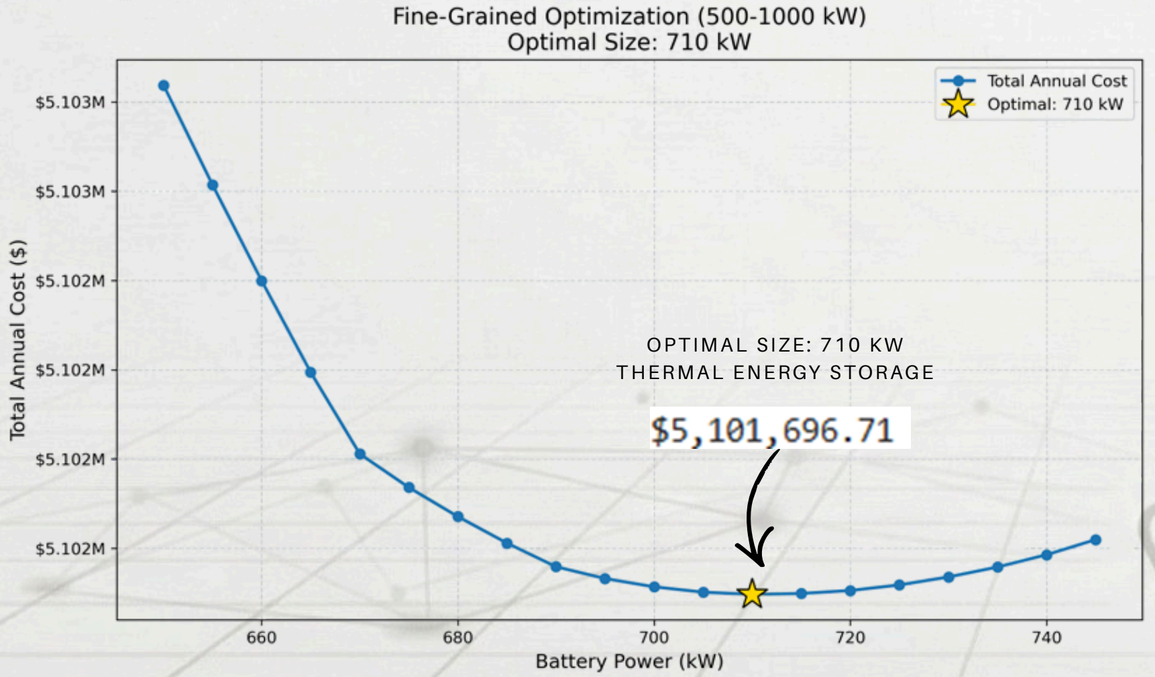
Optimal Size:
710 kW

Total Annual Cost:
$5.101M

(vs. Baseline $5.21M) | Annual Savings: **~$110,139**







COST OF EFFICIENCY LOSS

LITHIUM ION BATTERY

For 1kWh discharging:

$1/(0.95 * 0.95)$

= 1.108 kWh charge

It takes 0.108 kWh in excess from the grid.

Cost = 0.108 * 0.13/kWh

= **0.01404 / kWh** of discharging

THERMAL ENERGY

For 1kWh discharging:

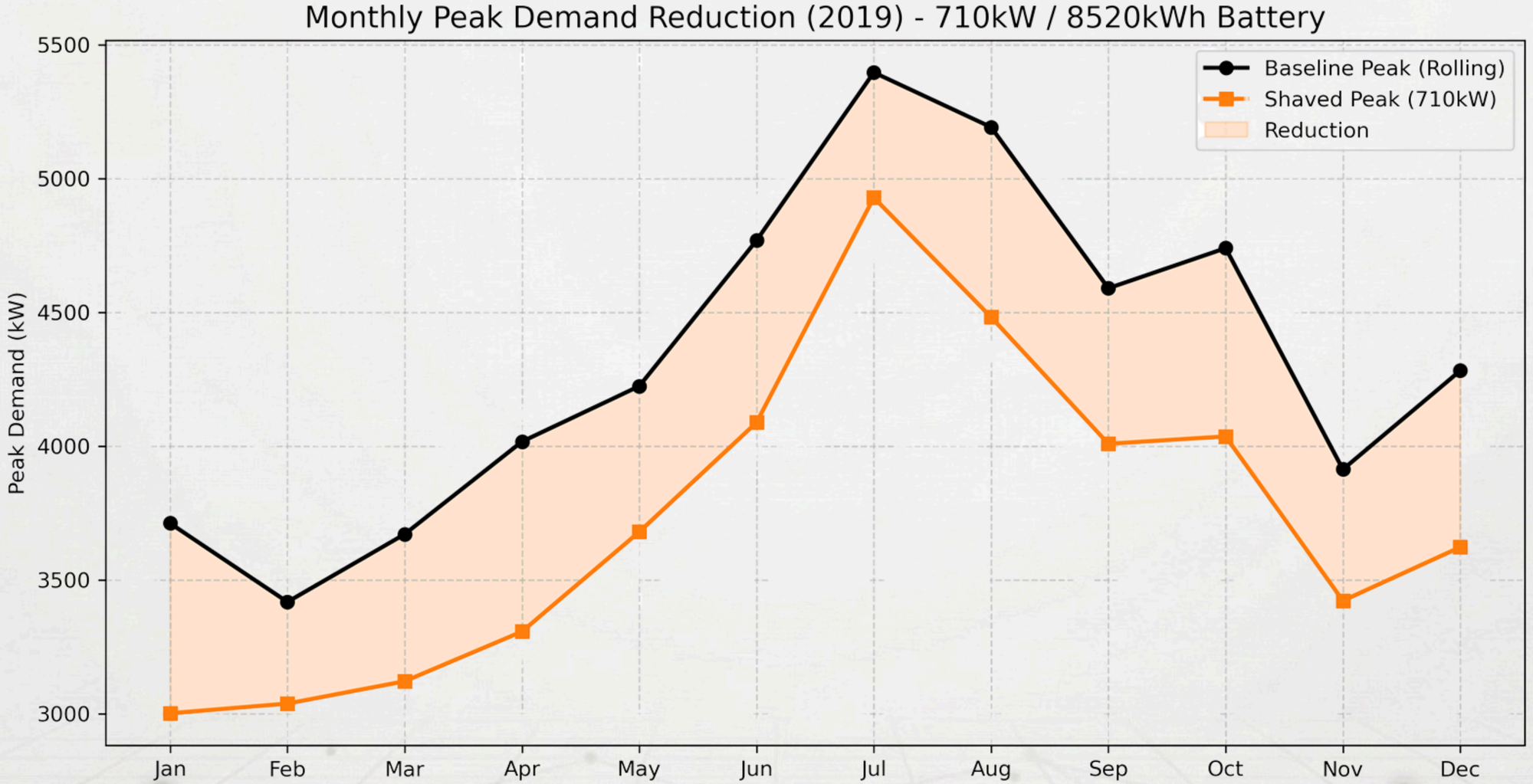
$1/(0.7 * 0.7)$

= 2.041 kWh charge

It takes 1.041 kWh in excess from the grid.

Cost = 1.041 * 0.13/kWh

= **0.135 / kWh** of discharging


Monthly Peak Demand Reduction (2019) - 710kW / 8520kWh Battery
Peak Demand (kW)
5500
5000
4500
4000
3500
3000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Baseline Peak (Rolling)
Shaved Peak (710kW)
Reduction

Stage 2: Demand Forecasting (Machine Learning) - Model Selection & Performance

Methodology & Feature Engineering

1. Data Preparation (2018 Data)

2. Feature Extraction (lag_96, lag_4, lag_1, Month, Hour, Weekend)

3. Model Training & Validation

Models Tested

- Neural Network
- Random Forest
- Linear Regression

Model Performance (RMSE Comparison)

Model	RMSE (kW)
Neural Network	60.70 kW
Random Forest	63.51 kW
Linear Regression (Winner)	**59.12 kW**

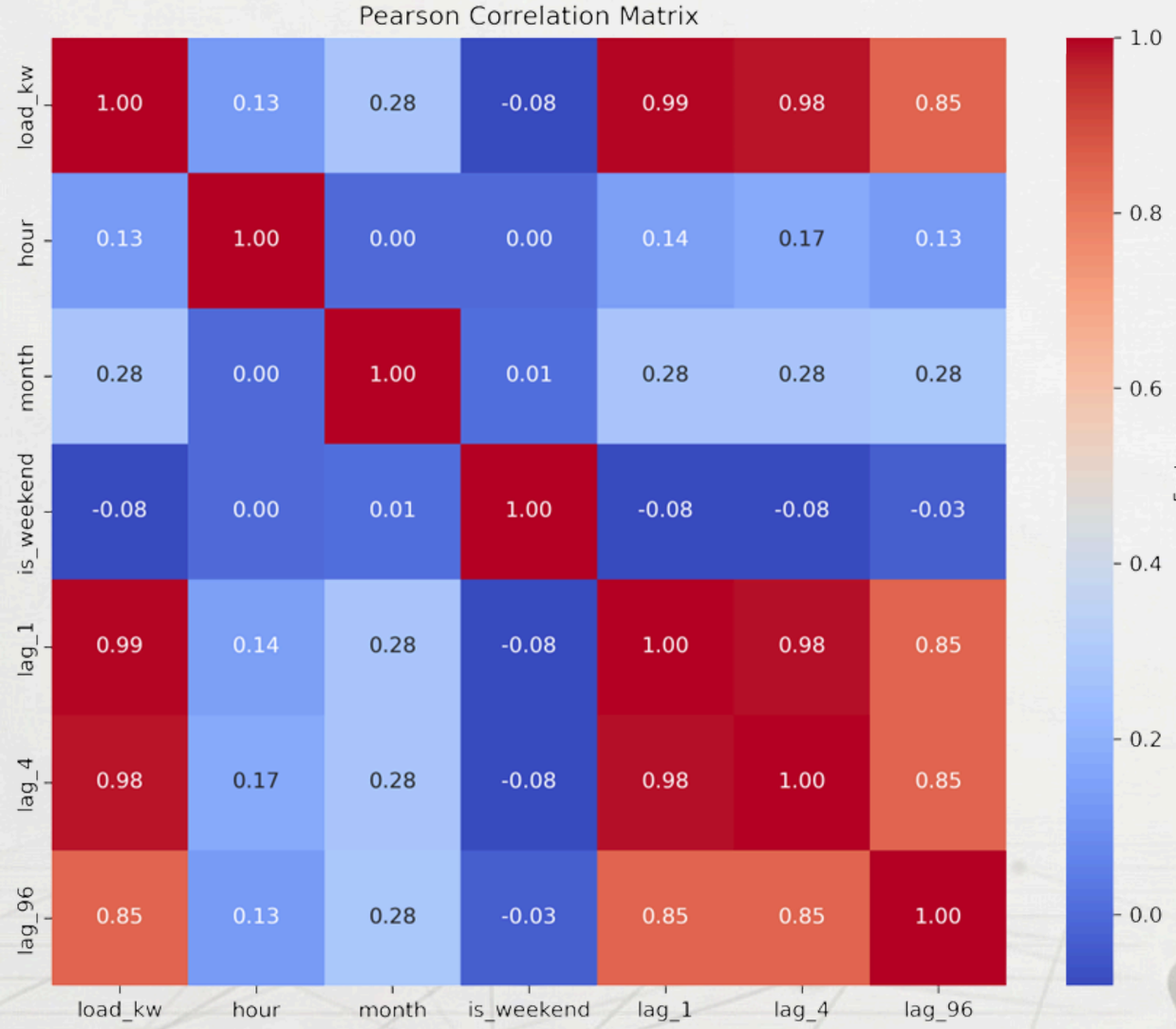
Why Linear Regression Won

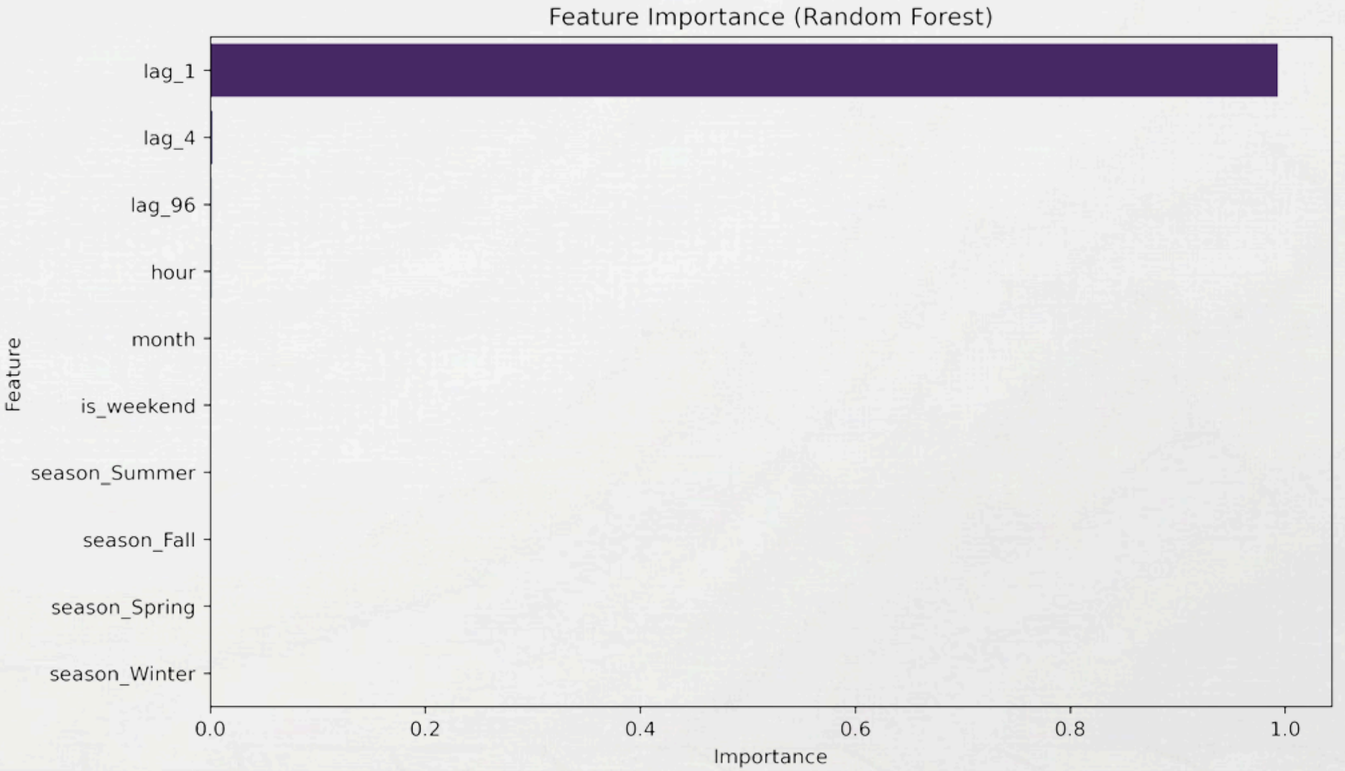
Structured Campus Pattern:
Campus load follows highly predictable, schedule-based patterns

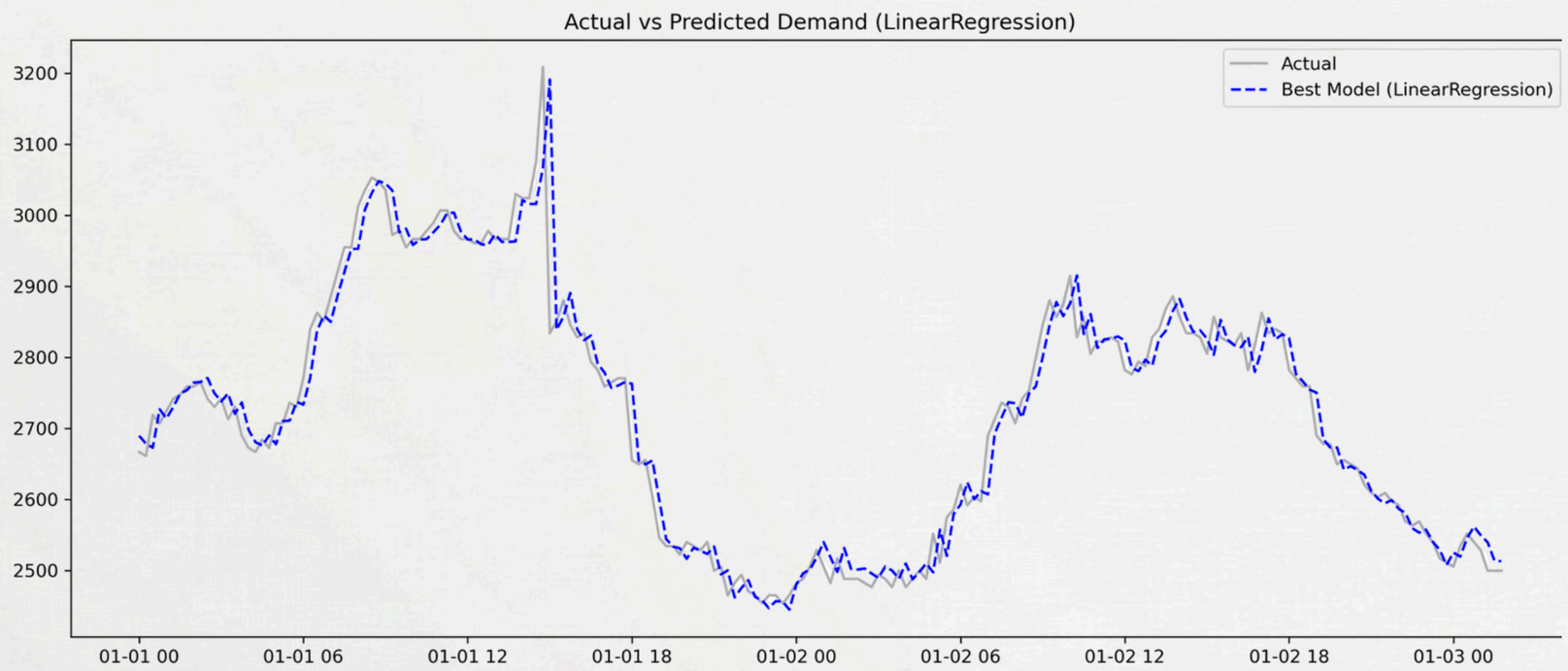
Linear Regression Efficacy:
Linear Regression Efficacy Effectively captured strong linear correlations with time-lagged features, avoiding overfitting seen in complex models.

Linear Regression selected as the optimal forecast engine for real-world dispatch simulation.

Correlation Matrix & Feature Importance - Linear Regression

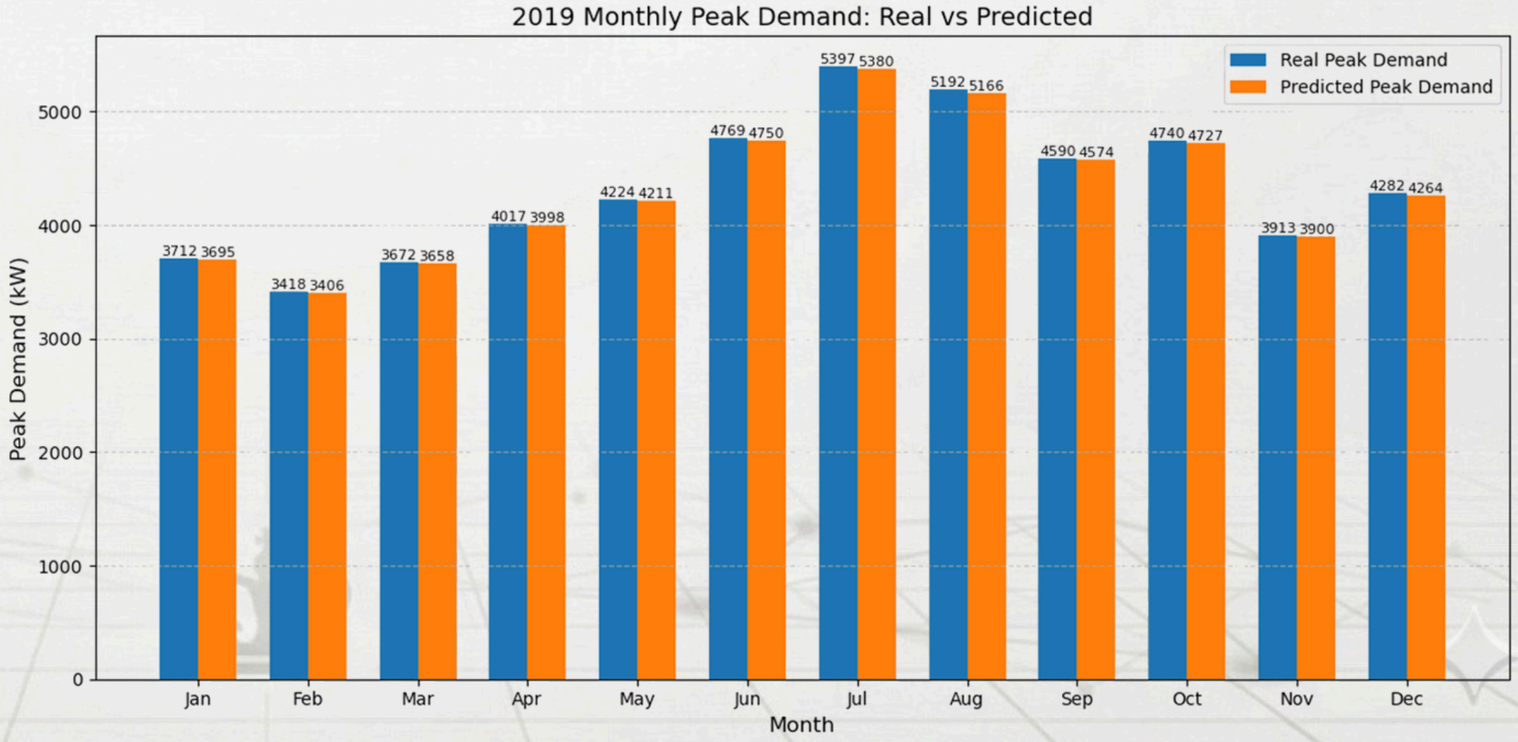




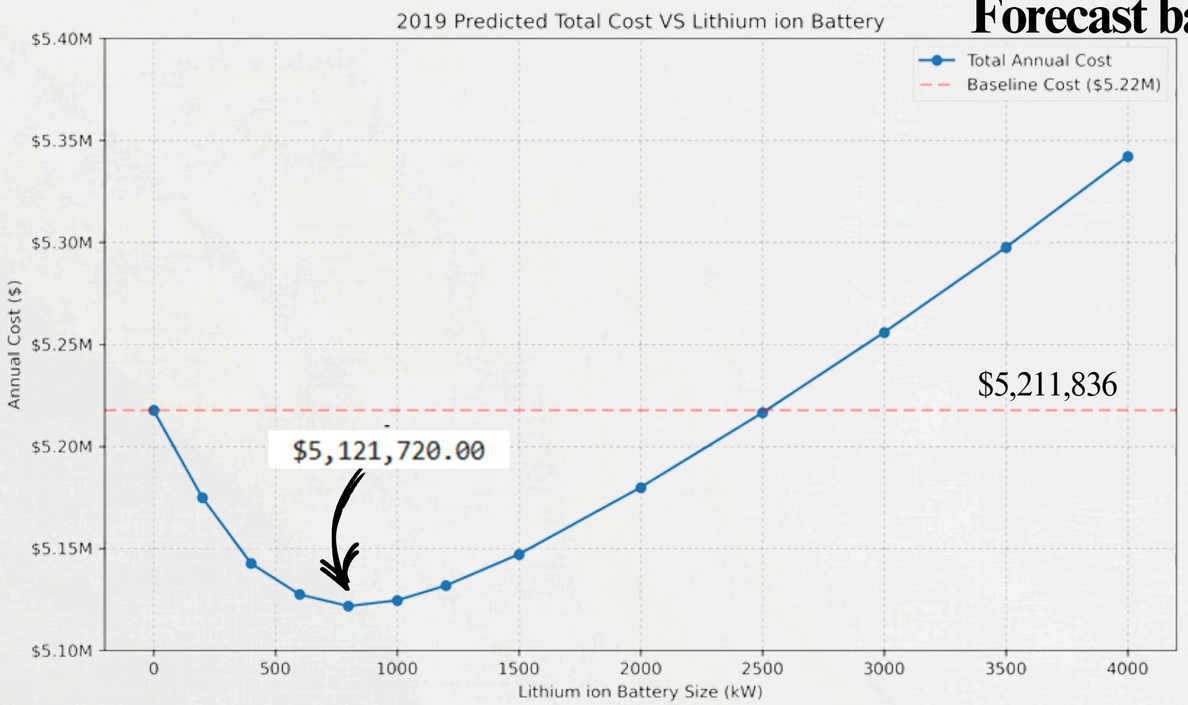


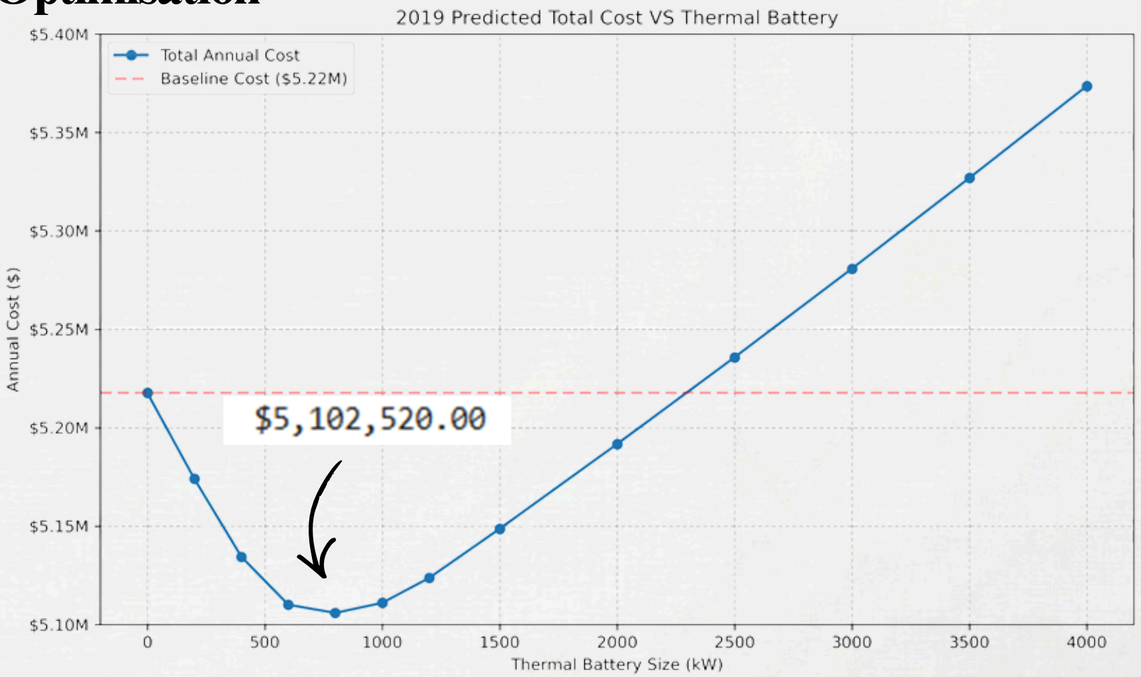
The Linear Regression model (blue) demonstrates high performance in tracking Actual Demand (grey), accurately capturing daily peaks and troughs.

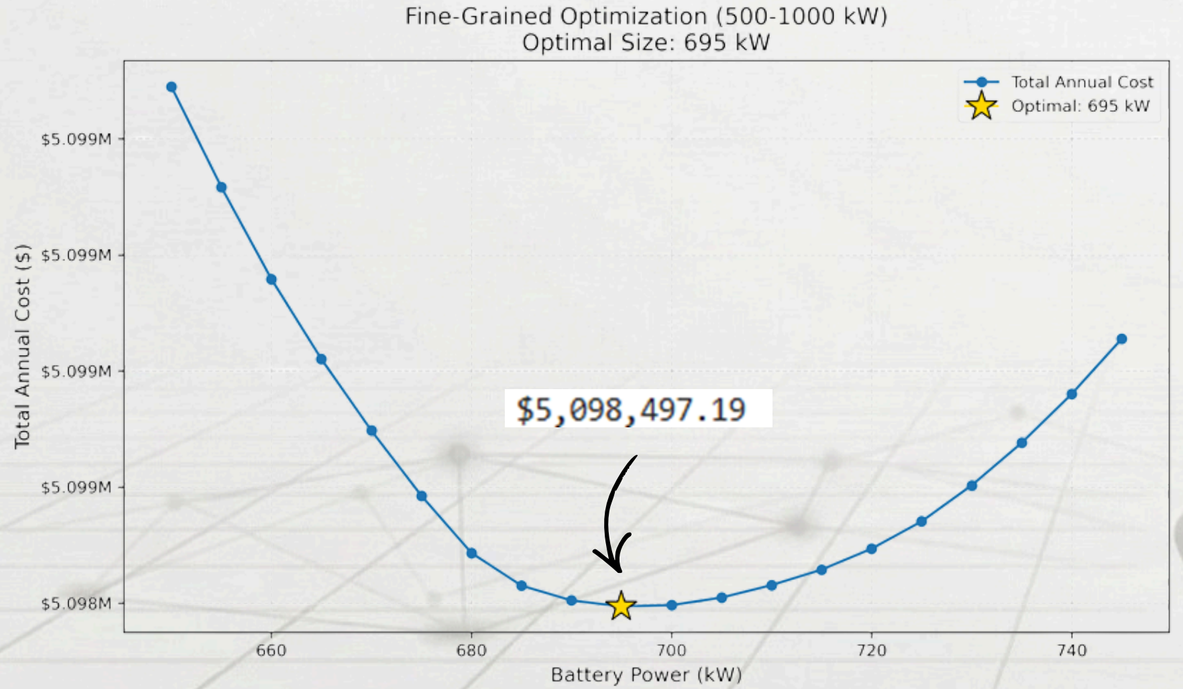
Peak Accuracy Example: July Peak: Real 5397 kW vs. Predicted 5380 kW. This minimal error validates model robustness for operational use



Forecast based Optimisation







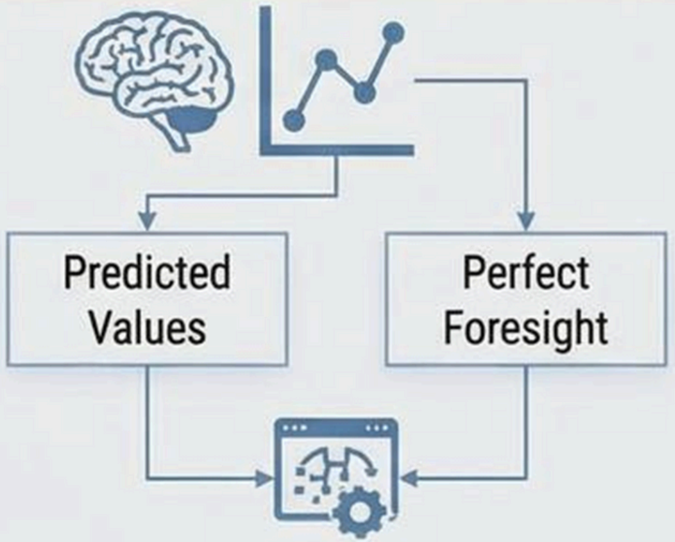
Optimal Size: 695 kW

Thermal Energy Storage

Stage 3: Optimization under Uncertainty & Real-World Simulation

Re-optimization using Predicted Demand to Account for Forecast Errors and Hedge Risk

Simulation Approach



System Optimization Model

Key Results & Risk Hedging

New Optimal Size: **695 kW**

(Downsized from 710 kW to Mitigate Risk)

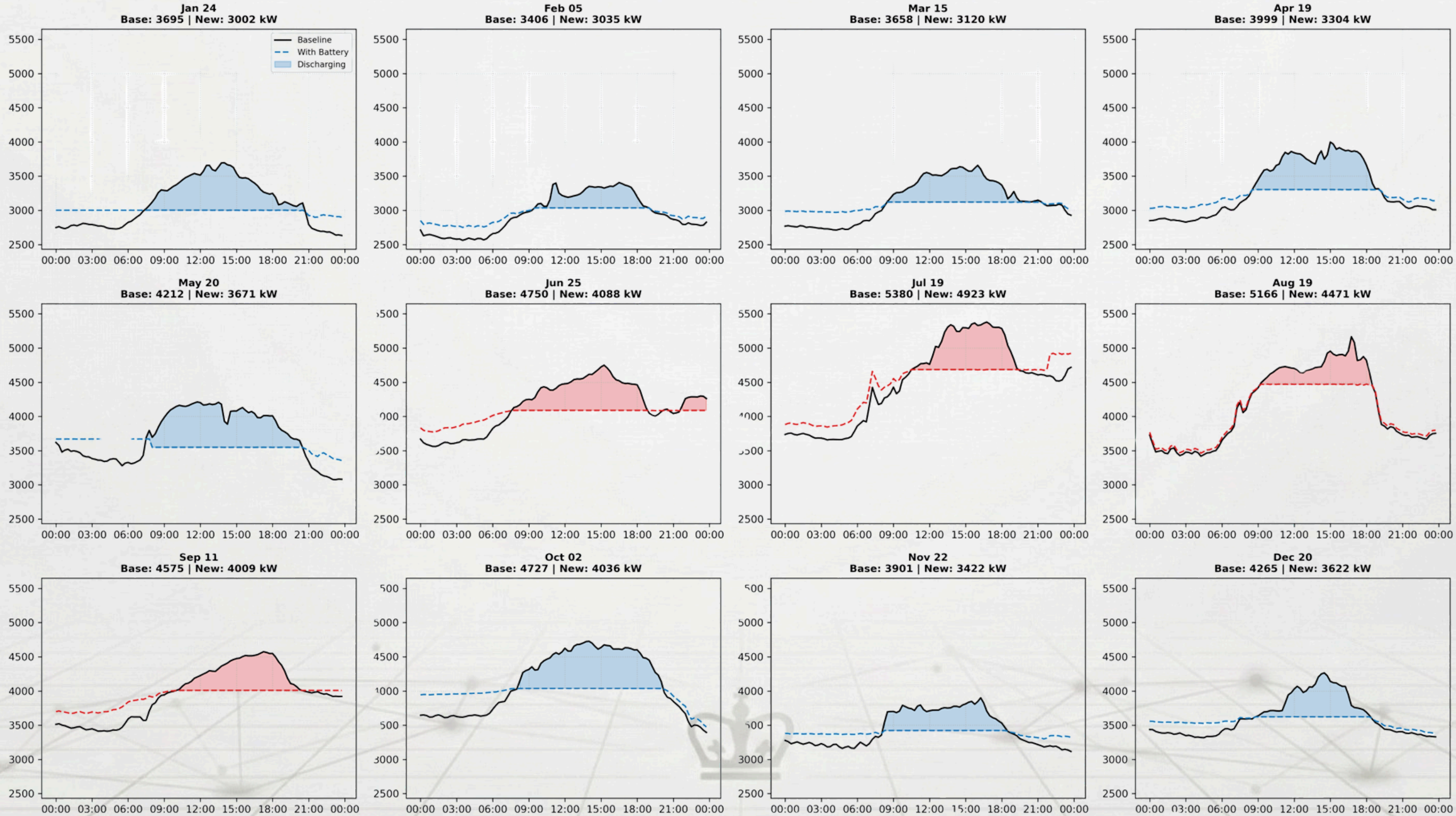
Cost of Forecast Uncertainty: **$3,200**

Key Insight & Robustness

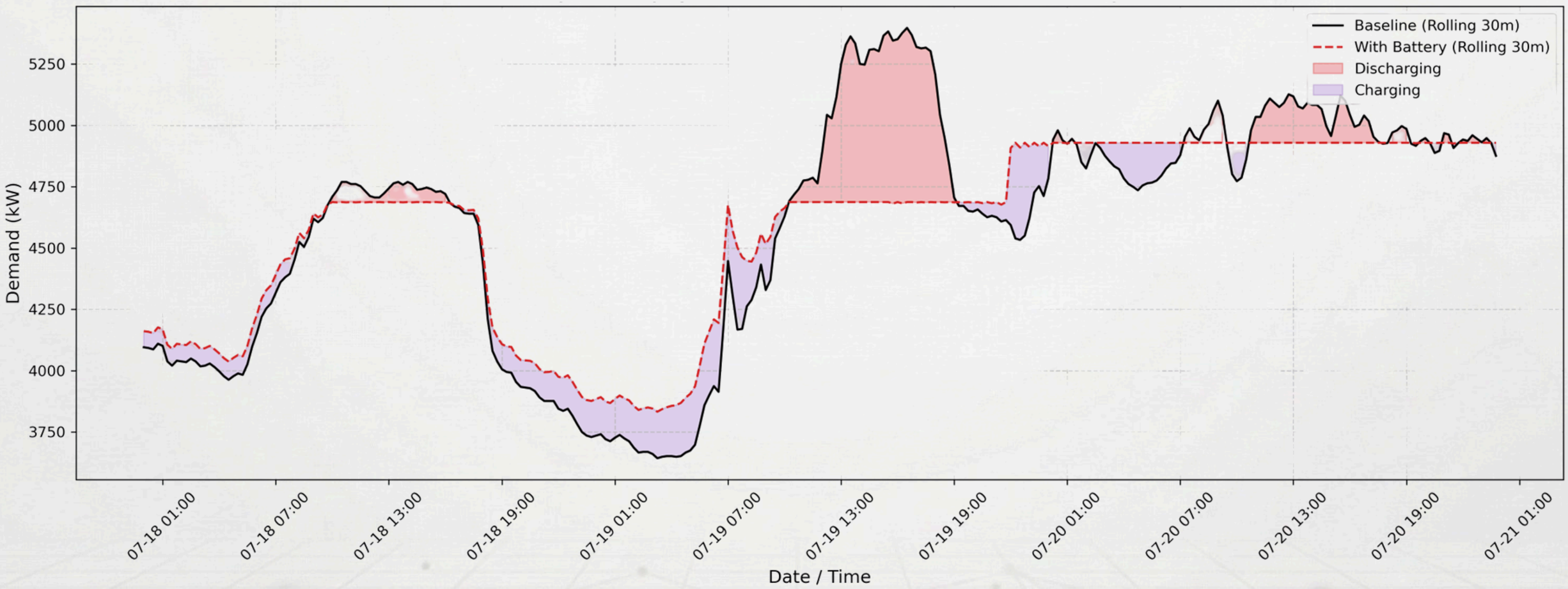
Negligible financial difference between Perfect Foresight and ML Model confirms exceptionally robust forecasting.

Uncertainty has minimal impact on peak-shaving savings.

Peak Shaving Profiles (2019): 695kW Battery | 3x4 Grid
Red = Summer | Blue = Winter



Peak Shaving Profile - July



Financial Analysis: NPV & Viability

Cash Flow Summary

	CapEx (Initial Investment)	$764,500
	Annual Savings	~$110,139
%	Discount Rate	8%
	Total Net Present Value (NPV)	**-$25,458**

Viability Assessment

Marginally Unviable Today

Project covers 97% of capital costs. Technically sound, but financial viability is marginal under current CapEx.

Path to Positive NPV:
A 3.5% CapEx Reduction or Incentive Turns Project Positive.

Conclusion & Recommendations

Technical Conclusion

Optimal Tech: Thermal Energy Storage is superior to Lithium-Ion for this application.

Optimal Size: 695 kW is the recommended capacity.

Forecasting: Linear Regression is the most effective tool for campus load prediction (RMSE ~59kW).

Strategic Recommendation

Although the current NPV is -$25k, the project is a strong candidate for implementation if:

CapEx Reduction: A decrease in battery costs of just ~3.5% would turn the NPV positive.

Incentives: Securing sustainability grants or demand response incentives would easily bridge the $25k gap.

Overall Assessment: Technically sound, financially borderline, highly sensitive to small cost changes.

tusind tak
谢谢
dakujem vám
ありがとう
ngiyabonga
dziękuję
merci
thank
you
suksema
danke
baie dankie
धन्यवाद
molte grazie
gracias
obrigada
takk
obrigado
gràcies
dank u
teşekkür ederim
شكرا
tänan
tack så mycket
teşekkür ederim
mahalo

Appendix

Year	Cash Flow	NPV
0	-$764,500.00	-764500.00
1	$110,139.00	101980.56
2	$110,139.00	94426.44
3	$110,139.00	87431.89
4	$110,139.00	80955.45
5	$110,139.00	74958.75
6	$110,139.00	69406.25
7	$110,139.00	64265.05
8	$110,139.00	59504.67
9	$110,139.00	55096.92
10	$110,139.00	51015.67
	Total NPV	-25458.34

Negative NPV indicating that battery integration is not financially viable